Undergraduate Topics in Mathematics 5
Abstract Algebra

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Contents

Combinatorics

1.1 Introduction

- **What is combinatorics?**: Combinatorics is a collection of techniques and a language for the study of finite or countably infinite discrete structures. Given a set of elements and possibly some structure on that set, typical questions are
  - Does a specific arrangement of the elements exists?
  - How many such arrangemets are there?
  - What properties do these arrangements have?
  - Which one of the arrangemetns is maximal, minimal, or optimal according to some criterion?
- **Counting the number of subsets for a set**: Let $[n] = \{1, 2, ..., n\}$, and let $f(n)$ be the number of subsets of $[n]$. Then $f(n) = 2^n$. For any particular subset of $[n]$, each element is either in that subset or not. Thus, to construct a subset, we have to make one of two choices for each element of $[n]$. Furthermore, these choices are independent of each other. Hence, the total number of choices, and consequently the total number of subsets is

$$\underbrace{2 \times 2 \times ... \times 2}_{n} = 2^n.$$

- **Number of subsets without consecutive integers**: For a sequence $[n] = \{1, ..., n\}$ we can count the number of subsets given by $f(n)$, that do not contain consecutive integers with the recurrence relation

$$f(n) = f(n-1) + f(n-2).$$

  We consider two cases

  1. n in not included in the subsets
  2. n is included in the subsets. In this case, we build the subsets considering the subsequence $[n-2] = \{1, ..., n-2\}$. Note that if we include n, we must exclude $n-1$, because $n-1$ and n are consecutive, this will become cleare in the upcoming example.

  Consider the sequence $[n] = \{1, 2, 3, 4\}$. By the relation above,

$$f(4) = f(3) + f(2).$$

  Before we are able to compute this, we must define our base cases.

$$f(n) = \begin{cases} 3 & \text{if } n = 2 \\ 2 & \text{if } n = 1 \end{cases}.$$

  If $n = 2$, we have $\{1, 2\}$, and the allowed subsets are $\varnothing, \{1\}, \{2\}$. If we have $n = 1$, the subsets are $\{\varnothing, \{1\}\}$. Thus

$$\begin{aligned} f(4) &= f(3) + f(2) = f(2) + f(1) + f(2) \\ &= 3 + 2 + 3 = 8. \end{aligned}$$

Let's explicitly break up the given sequence so we can see whats going on. In the first case, n is excluded, thus the sequence becomes $\{1,2,3\}$. If n is included, the sequence becomes $\{1,2\}$, where we build the subsets of $\{1,2\}$, and then add 4 to each one. Thus,

$$\begin{aligned}\{1,2,3\}+\{1,2\} &= \{1,2,3\}+\varnothing+\{1\}+\{2\}\\ &= \{1,2,3\}+\{4\}+\{1,4\}+\{2,4\}.\end{aligned}$$

Since the sequence $\{1,2,3\}$ in not a base case, we must split this one up aswell, we have

$$\begin{aligned}\{1,2,3\} &= \{1,2\}+\{1\}+\{4\}+\{1,4\}+\{2,4\}\\ &= \varnothing+\{1\}+\{2\}+\varnothing+\{1\}+\{4\}+\{1,4\}+\{2,4\}\\ &= \varnothing+\{1\}+\{2\}+\{3\}+\{1,3\}+\{4\}+\{1,4\}+\{2,4\}.\end{aligned}$$

Thus, we conclude all "good" subsets of $[n]$ either have n or don't have n. The ones that don't have n are exactly the "good" subsets of $[n-1]$. The "good" subsets of $[n]$ that include n are exactly the "good" subsets of $[n-2]$ together with n. Thus $f(n) = f(n-1) + f(n-2)$ $\blacksquare$

1.2 Induction and recurrence relations

- **Principal of Mathematical Induction**: Given an infinite sequence of propositions

$$P_1, P_2, P_3, ...P_n, ...,.$$

In order to prove that all of them are true, it is enough to show two things

1. **The base case:** P_1 is true
2. **The inductive step**: For all positive integers k, if P_k is true, then so is P_{k+1}

Example: Show that

$$1 + 2 + 3 + ... + n = \frac{n(n+1)}{2}.$$

Base case:

$$1 = \frac{1(1+1)}{2} = \frac{2}{2} = 1.$$

Inductive step: P_k is given by

$$1 + 2 + 3 + ... + k = \frac{k(k+1)}{2}.$$

P_{k+1} is given by

$$1 + 2 + 3 + ... + k + k + 1 = \frac{k + 1(k+2)}{2}.$$

If $1 + 2 + 3 + ... + k = \frac{k(k+1)}{2}$, then

$$\begin{aligned} 1 + 2 + 3 + ... + k + k + 1 &= \frac{k + 1(k+2)}{2} \\ \frac{k(k+1)}{2} + k + 1 &= \frac{k + 1(k+2)}{2} \\ \frac{k(k+1) + 2k + 2}{2} &= \frac{k^2 + 3k + 2}{2} \\ \frac{k^2 + 3k + 2}{2} &= \frac{k^2 + 3k + 2}{2}. \end{aligned}$$

Thus, we have showed that $P_k \implies P_{k+1}$ ■.

Note: Our aim is not to directly prove P_{k+1}, but to prove that P_k implies P_{k+1}. In the inductive step we assume P_k to be true, then show under this assumption, P_{k+1} is also true.

- **Understanding gauss's formula for the sum of the first n natural numbers**: Suppose we want to find the sum $1+2+3+...+(n-1)+n$. We could have discovered the formula that we proved above by first writing the sum twice

$$\begin{aligned} &1 + 2 + 3 + ... + (n-1) + n \\ &n + (n-1) + (n-2) + ... + 2 + 1. \end{aligned}$$

The sum of the two numbres in each column is $n+1$, and there are n columns, so the total sum is $n(n+1)$, it then follows that the actual sum is $\frac{1}{2}n(n+1)$

- **Trianglular numbers**: The sequence of integers

$$\begin{aligned}
&1 && 3 = 1 + 2 \\
&6 = 1 + 2 + 3 \\
&10 = 1 + 2 + 3 + 4 \\
&15 = 1 + 2 + 3 + 4 + 5 \\
&....
\end{aligned}$$

  Are called *triangular numbers*. If you were to make a triangle of dots out of the sum, where the highest number is the base, the second highest is the layer ontop of the base, etc, you would form a triangle.

- **Strong induction**: Given an infinite sequence of propositions

$$P_1, P_2, P_3, ..., P_n.$$

  In order to demonstrate that all of them are true, it is enough to know two things.

  1. **The base case**: P_1 is true
  2. **The inductive step**: For all integers $k \geqslant 1$, if $P_1, P_2, P_3, ..., P_k$ are true, then so is P_{k+1}

- **Pingala-fibonacci numbers**: Define a sequence of positive integers as follows: $F_0 = 0, F_1 = 1$, and for $n = 2, 3, ...$ we have

$$F_n = F_{n-2} + F_{n-1}.$$

  This sequence is also known as *the fibonacci sequence*.

- **Lucas numbers**: Change the initial values on the fibonacci sequence. Let $L_0 = 2, L_1 = 1$, and $L_n = L_{n-2} + L_{n-1}$. Then, we get the *Lucas numbers*

$$2, 1, 3, 4, 7, 11, 18, 29, 47, ...$$

$$\mathcal{L}.$$

The Fourier Transform

- **The Fourier Transform**: The Fourier transform is a tool that rewrites a function in terms of its frequency content. Instead of describing a signal by "what value it has at each time," the Fourier transform describes it by "how much of each frequency is present."

  For a function $f(t)$, the FT is written as

  $$\hat{f}(\omega) = \mathcal{F}\left\{f(t)\right\},$$

  and defined by

  $$\hat{f}(\omega) = \int_{-\infty}^{\infty} e^{-i\omega t} f(t)\, dt.$$

  With inverse

  $$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \hat{f}(\omega) e^{i\omega t}\, d\omega,$$

  which recovers the original function $f(t)$. Note that ω here is the angular frequency.

- **Derivatives become multiplication**: One reason Fourier transforms are so powerful is that they turn differentiation into multiplication.

  $$\mathcal{F}\left\{f'(t)\right\} = i\omega \hat{f}(\omega).$$

Abstract Algebra

3.1 Relations and functions

- **Relations**: Let S and T be sets. A **relation** from S to T is a subset ρ of $S \times T$, where

$$S \times T = \{(s,t) : s \in S, t \in T\}.$$

If $s \in S$ and $t \in T$, then we write $s\rho t$ if $(s,t) \in \rho$, otherwise we write $s\not\rho t$. In particular, a relation on S is a relation from S to S.

- **Properties of relations**: Let ρ be a relation on S,
  - **Reflexive**: ρ is reflexive if $a\rho a$ for all $a \in S$
  - **Symmetric**: ρ is symmetric if $a\rho b$ implies $b\rho a$
  - **Transitive**: ρ is transitive if $a\rho b$ and $b\rho c$ implies that $a\rho c$

  Note: These properties are defined only for relations from a set to itself.

- **Equivalence relation**: An **equivalence relation** on a set S is a relation that is reflexive, symmetric, and transitive. We use $\sim$ to denote an equivalence relation.

  As an example, on $\mathbb{R}$, let us say that $a \sim b$ if and only if $a + b$ is even. First, consider $a, b \in \mathbb{R}$. Observe that

$$a \sim a$$

since whether or not a is even or odd, the sum is always even. So, $\sim$ is reflexive. Next, we notice that if we assume $a \sim b$ (so $a + b$ is even), it immediately follows that $b + a$ is even. So, $b \sim a$ and $\sim$ is symmetric.

Now, let $a, b, c \in \mathbb{R}$. Assume that

$$a + b, \ b + c$$

are both even. Hence,

$$a + b + b + c = a + 2b + c = a + c + 2b$$

is also even. Notice that $2b$ is even, so we consider two cases, either $a + c$ is odd and $\sim$ is not transitive or $a + c$ is even. If $a + c$ is odd, then we have $a + c = 2\ell + 1$, and

$$a + c + 2b = 2\ell + 1 + 2b = 2(\ell + b) + 1,$$

which is odd. But, we said that $a + c + 2b$ is even, a contradiction. Thus, $a \sim c$ and $\sim$ is transitive.

Since $\sim$ is reflexive, symmetric, and transitive, it is an equivalence relation. ■

- **Equivalence class**: Let $\sim$ be an equivalence relation on a set S. If $a \in S$, then the **equivalence class** of a, denoted $[a]$, is the set $\{b \in S : a \sim b\}$

- **Partition**: Let S be a set, and let T be a set of nonempty subsets of S. We say that T is a **partition** of S if every $a \in S$ lies in exactly one set in T.

- **Theorem 1.1**: Let S be a set and $\sim$ be an equivalence relation on S. Then the equivalence classes with respect to $\sim$ form a partition of S. In particular, if $a \in S$, then $a \in [a]$, and $a \in [b]$ if and only if $[a] = [b]$.

Proof. Assume S is a set, and $\sim$ is an equivalence relation on S. To show that the equivalence relations with respect to $\sim$ form a partition of S, we must show that the equivalence relations are a set of nonempty pairwise disjoint subsets of S.

Let $a \in S$. By reflexivity of $\sim$, $a \sim a$. Thus, $a \in [a]$ and the equivalence classes are nonempty. Now, assume that

$$d \in [a] \cap [b].$$

Thus,

$$a \sim d, \quad b \sim d.$$

Now, if $c \in [a]$, then $a \sim c$. Notice that by transitivity,

$$d \sim a,\ a \sim c \implies c \sim d.$$

Similarly,

$$c \sim d,\ d \sim b \implies c \sim b.$$

Thus, $c \in [b]$. Since $c \in [a]$ implies that $c \in [b]$,

$$[a] \subseteq [b].$$

A similar argument, except starting with $c \in [b]$ yields that

$$[b] \subseteq [a].$$

Hence, if $d \in [a] \cap [b]$, it follows that $[a] = [b]$.

Therefore, if $a \in S$, then $a \in [a]$, and if $a \in [b]$, then $[b] = [a]$. Conversely, if $[a] = [b]$, then $a \in [a] = [b]$. ■

- **The partition**: So, now we know that for a equivalence relation $\sim$ on S,

  $$\sim \subseteq S \times S,$$

  if $a \in S$, then the equivalence class of a is the set

  $$[a] = \{b \in S :\ a \sim b\}.$$

  Further, $b \in [b]$, and so it must be that $[b] = [a]$. In other words, the representative element is not unique for each class.

- **Functions**: Let S and T be any sets. A **function**

  $$\alpha :\ S \to T$$

  is a rule assigning to each $s \in S$ an element $\alpha(s) \in T$.

- **Requirements of a function**: For a relation $f \subseteq A \times B$ to be called a **function** from A to B, it must satisfy

  1. **Existence / totality**: For all $a \in A$, there exists a $b \in B$ such that $(a, b) \in f$

     $$\forall a \in A,\ \exists b \in B\ :\ (a, b) \in f.$$

2. **Uniqueness / well-definedness**: Each input in A has exactly one output in B

$$\forall a \in A,\ \forall b_1, b_2 \in B,\ ((a, b_1) \in f \wedge (a, b_2) \in f) \implies b_1 = b_2.$$

- **Properties of functions**: Consider a function $\alpha :\ S \to T$,
  - **Injective**: α is injective if $\alpha(s_1) = \alpha(s_2)$ implies $s_1 = s_2$ for all $s_1, s_2 \in S$
  - **Surjective**: α is surjective if for all $t \in T$, there exits an element $s \in S$ such that $\alpha(s) = t$
  - **Bijective**: α is bijective (invertible) if it is both injective and surjective

  We call a bijective function a **bijection**, which is therefore also a **injection** and a **surjection**.

  Notice that the inverse of a function only satisfies existence / uniqueness if it is both injective (uniqueness) and surjective (existence).

- **Composition**: Let R, S, and T be sets, and let $\alpha :\ R \to S$ and $\beta :\ S \to T$ be functions. Then, the **composition** $\beta \circ \alpha$, or simply $\beta\alpha$, is the function from R to T given by

$$(\beta\alpha)(r) = \beta(\alpha(r))$$

  for all $r \in R$. Note that $\alpha\beta$ and $\beta\alpha$ are likely different functions.

- **Theorem 1.2: Properties of the composition**: Let $\alpha :\ R \to S$, $\beta :\ S \to T$, and $\gamma :\ T \to U$ be functions. Then,
  1. $(\gamma\beta)\alpha = \gamma(\beta\alpha)$
  2. If α and β are injective, then so is $\beta\alpha$
  3. If α and β are surjective, then so is $\beta\alpha$
  4. If α and β are bijective, then so is $\beta\alpha$

  Proof. First, notice that

$$(\gamma\beta)\alpha = \gamma(\beta(\alpha)),$$
$$\gamma(\beta\alpha) = \gamma(\beta(\alpha)).$$

  Hence, the first property holds. Now, consider $r_1, r_2 \in R$, and assume that

$$\beta(\alpha(r_1)) = \beta(\alpha(r_2)).$$

  Since β is injective, $\alpha(r_1) = \alpha(r_2)$, and since α is injective, $r_1 = r_2$. Thus, $\beta\alpha$ is injective.

  Next, consider $t \in T$. Since β is surjective, there exists a $s \in S$ such that $\beta(s) = t$. But, α is surjective, so there exists an $r \in R$ such that $\alpha(r) = s$. Thus,

$$(\beta\alpha)r = \beta(\alpha(r)) = \beta(s) = t.$$

  Since we showed that $\beta\alpha$ is both injective and surjective, it follows that $\beta\alpha$ is bijective. ■

- **Theorem 1.3: Existence of the inverse**: Let $\alpha :\ S \to T$ be a bijection. Then, there exists a bijection $\beta :\ T \to S$ such that $(\beta\alpha)(s) = s$ for all $s \in S$ and $(\alpha\beta)(t) = t$ for all $t \in T$.

Proof. Since α is bijective, for all $t \in T$, there exist a unique $s \in S$ such that $\alpha(s) = t$. Now, define $\beta : T \to S$ such that for that unique s that maps to t,

$$\beta(t) = s.$$

Then,

$$(\beta\alpha)(s) = \beta(\alpha(s)) = \beta(t) = s.$$

Now, for any $t \in T$, choose $s \in S$ such that $\alpha(s) = t$. By that same property of β,

$$\beta(t) = s,$$

so

$$(\alpha\beta)(t) = \alpha(\beta(t)) = \alpha(s) = t.$$

Now, we just need to show that β is a bijection. First, assume that $\beta(t_1) = \beta(t_2)$. Then,

$$\alpha(\beta(t_1)) = t_1,$$
$$\alpha(\beta(t_2)) = t_2.$$

But, $\beta(t_1) = \beta(t_2)$, so

$$\alpha(\beta(t_2)) = t_2 = \alpha(\beta(t_1)) = t_1.$$

Thus, β is injective. Next, $s \in S$, we have that

$$\beta(\alpha(s)) = s.$$

Thus, there exists a $t \in T$ such that $\beta(t) = s$ $(t = \alpha(s))$, so β is surjective. $\blacksquare$.

- **Permutation**: A permutation of a set S is a bijection from S to S.
- **Binary operation**: Let S be a set. Then, a **binary operation** on S is a function from $S \times S$ to S.

3.2 The Integers and Modular Arithmetic

3.2.1 Induction

- **Property 2.1: Well ordering axiom**: If S is a nonempty set of positive integers, then S has a smallest element.
- **Theorem 2.1: Mathematical induction**: Suppose that for each positive integer n, we have a proposition $P(n)$. Further suppose that

  1. $P(1)$ is true, and
  2. for each $n \in \mathbb{N}$, if $P(n)$ is true, then so is $P(n+1)$

  Then, $P(n)$ is true for every positive integer n.

  Proof. We proceed by contradiction. Recall that to show a statement $P \implies Q$ is true, it is merely sufficient to show that $P \wedge \neg Q$ is impossible. Thus, if we assume $\neg Q$, and yield a contradiction, $P \wedge \neg Q$ cannot occur and thus the statement is true.

  So, we assume that $P(n)$ is not true for every positive integer n. Thus, there is a nonempty set $S \subseteq \mathbb{N}$ such that for every element $n \in S$, $P(n)$ is false. By the well-ordering axiom, there exists a smallest element in S, call this element k.

  Observe that the statement assumes that $P(1)$ is true, and for each $n \in \mathbb{N}$, if $P(n)$ is true, then so is $P(n+1)$. So, $k > 1$. Further,

  $$k - 1 \notin S,$$

  since if $k-1 \in S$, $k-1+1 = k \notin S$, but k is indeed a member of S. However, since $k-1 \notin S$, $P(k-1)$ is true, and thus $P(k-1+1) = P(k)$ is true, a contradiction.

  Therefore, if $P(1)$ is true, and for each $n \in \mathbb{N}$, if $P(n)$ is true, $P(n+1)$ is also true, then $P(n)$ is true for every positive integer n. ∎

  Let's consider an example. Suppose we claim that for every positive integer n,

  $$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

  First, we observe that

  $$1^2 = \frac{1(1+1)(2(1)+1)}{6} = 1.$$

  So, $P(1)$ is true. Next, assume that $P(n)$ is true, where

  $$P(n) = 1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

  Now, observe that

  $$\begin{aligned} P(n+1) &= 1^2 + 2^2 + \cdots + n^2 + (n+1)^2 = P(n) + (n+1)^2 \\ &= \frac{n(n+1)(2n+1)}{6} + (n+1)^2 = \frac{n(2n+1)}{6} + (n+1) \\ &= \frac{n(2n+1) + 6(n+1)}{6} = \frac{2n^2 + n + 6n + 6}{6} = \frac{2n^2 + 7n + 6}{6}. \end{aligned}$$

From here, we can get that

$$\frac{2n^2+7n+6}{6} = \frac{(n+2)(2n+3)}{6}.$$

So,

$$\frac{P(n)}{n+1} + (n+1) = \frac{(n+2)(2n+3)}{6}.$$

Thus,

$$P(n) + (n+1)^2 = P(n+1) = \frac{(n+1)((n+1)+1)(2(n+1)+1)}{6}$$

Thus, $P(n)$ implies $P(n+1)$, and P therefore holds for every positive integer n.

- **Binomial theorem**: Let a and b be real numbers, and n a positive integer. Then,

$$\begin{aligned}(a+b)^n &= a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \cdots + \binom{n}{n-1}ab^{n-1} + b^n \\ &= \sum_{k=0}^{n-1}\binom{n}{k}a^{n-k}b^k.\end{aligned}$$

Proof. *prove by induction on n.*

- **Strong induction**: Suppose that, for each positive integer n, we have a proposition $P(n)$. Further suppose that

  1. $P(1)$ is true, and
  2. for each integer $n > 1$, if $P(k)$ is true for every $k < n$, then $P(n)$ is true.

  Then, $P(n)$ is true for every positive integer n.

Proof. Assume that the two conditions hold, but $P(n)$ is not true for every positive integer n. Then, let S be the set of integers in which $P(s)$ is false. Since $S \subseteq \mathbb{N}$ and is nonempty, here is a smallest element j by the well-ordering axiom. Call this smallest element ℓ.

Note that since $P(1)$ is true, $\ell > 1$. Since ℓ is the smallest element in S, and therefore the first element that makes the proposition false, it must be that $P(k)$ is true for all $k < \ell$. But, that would imply that $P(\ell)$ is in fact true, a contradiction.

Thus, there is no such set of elements S that make P false, so $P(n)$ true for all $n \in \mathbb{N}$.

3.2.2 Some Number Theory

- **Theorem 2.4 (Division algorithm)**: Let $a, b \in \mathbb{Z}$ with $b > 0$. Then, there exists unique integers q and r such that

  $$a = bq + r, \quad 0 \leqslant r < b.$$

  We call q and r the **quotient** and **remainder** respectively.

  Proof. First, let

  $$S = \{a - bt :\ t \in \mathbb{Z},\ a - bt \geqslant 0\}.$$

  First, if $0 \in S$, then $a - bq = 0$ for some $q \in \mathbb{Z}$, and so

  $$a = bq + 0$$

  and the theorem holds. So, $S \subseteq \mathbb{N}$. We first claim that S is nonempty. Let

  $$t = -|a|.$$

  Then,

  $$a - bt = a + |a|\, b.$$

  If $a \geqslant 0$, then $a + |a|\, b \geqslant 0$ since $b > 0$. If $a \leqslant 0$ then

  $$a + |a|\, b = a - ab = a(1 - b).$$

  Notice that since $b > 0, b \in \mathbb{Z}$, it must be that $b \geqslant 1$. Thus,

  $$1 - b \leqslant 0.$$

  So, $a(1 - b) \geqslant 0$. Thus, we have shown the existence of at least one t such that $a - bt \in S$, and so S is nonempty.

  Since S is a nonempty subset of the natural numbers, we have a least element by the well-ordering axiom, say

  $$r = a - bq.$$

  Recall that $r \geqslant 0$ by definition. Suppose that $r \geqslant b$, then $0 \leqslant r - b < r$. But,

  $$r - b = a - bq + b = a - b(q + 1) < r,$$

  a contradiction, since r is the smallest element. Note that

  $$r - b = a - b(q + 1)$$

  is indeed a member of S, since $r - b \geqslant 0$. Thus,

  $$a = bq + r, \quad 0 \leqslant r < b.$$

  For uniqueness, suppose that

  $$a = bq_1 + r_1 = bq_2 + r_2$$

  with $q_i, r_i \in \mathbb{Z}$ and $0 \leqslant r_i < b$. Then,

  $$b(q_1 + q_2) = r_1 - r_2.$$

If we take the absolute value,

$$|b(q_1 + q_2)| = |r_1 - r_2|\,.$$

But, $b > 0$, so

$$b\,|q_1 - q_2|\,|r_2 - r_1|\,.$$

Notice that since $q_1, q_2 \in \mathbb{Z}$, $q_1 - q_2 \in \mathbb{Z}$, so if $q_1 \neq q_2$,

$$b\,|q_1 - q_2| \geqslant b.$$

But, $0 \leqslant r_1, r_2 < b$, so

$$|r_2 - r_1| < b,$$

and $|r_2 - r_1| = b\,|q_1 - q_2|$, which we just said was $\geqslant b$, a contradiction. Hence, $q_1 = q_2$. In that case,

$$b(q_1 - q_2) = b(0) = r_2 - r_2.$$

Thus, $r_1 = r_2$. ■.

- **Divides**: Let a and b be integers, we say that a divides b if there exists an integer c such that $b = ac$, in which case we write

  $$a \mid b.$$

  If no such c exists, we write $a \nmid b$

- **Lemma 2.1**: Let $a, b, c \in \mathbb{Z}$. Then,

  1. if $a \mid b$ and $b \mid c$, then $a \mid c$
  2. if $a \mid b$ and $b \neq 0$, then $a \leqslant |b|$
  3. if $a \mid b$ and $a \mid c$, then $a \mid (bu + cv)$ for any $u, v \in \mathbb{Z}$

Proof. First, assume that $a, b, c \in \mathbb{Z}$. (1). Assume that $a \mid b$, and $b \mid c$. Then,

$$b = a\ell, \quad c = bk.$$

If $k = 0$, then $c = 0$, and $a \mid c$. So, assume that $k \neq 0$. Then,

$$b = \frac{c}{k},$$

and so

$$\frac{c}{k} = a\ell.$$

Hence, $c = a(\ell k)$, and $a \mid c$.

(2) Assume that $a \mid b$, $b \neq 0$. So,

$$b = ar, \quad r \in \mathbb{Z}.$$

Consider the case that $a \leqslant |b|$ is false, so $|b| < a$, or

$$-a < b < a.$$

Notice that $a = 0$ would imply that $b = 0$, but $b \neq 0$. So, $a \neq 0$. Then,

$$-1 < \frac{b}{a} = r < 1.$$

But, this would contradict the fact that $r \in \mathbb{Z}$. So, $a \leqslant |b|$.

(3). Suppose that $a \mid b$ and $a \mid c$, so

$$b = ar,\ c = ak, \quad r, k \in \mathbb{Z}.$$

Let $u, v \in \mathbb{Z}$,

$$bu = aru,\ cv = akv.$$

Then,

$$bu + cv = aru + akv = a(ru + kv).$$

Since $ru + kv \in \mathbb{Z}$, $a \mid bu + cv$ for all $u, v \in \mathbb{Z}$. ■.

- **Greatest common divisor**: Let a and b be integers, not both zero. Then, the **greatest common divisor (gcd)** of a and b, written (a, b), is the largest positive integer g such that $g \mid a$ and $g \mid b$.

  Notice that (a, b) is always at least 1, and $(a, b) \leqslant |a|$. So, only numbers from 1 to $|a|$ are considered.

- **Lemma 2.2**: Take any integers a, b with $a \neq 0$. Then,

  1. $(a, b) = (-a, b)$
  2. $(a, 0) = |a|$

  Proof. Assume a, b are integers with $a \neq 0$.

  (1). Assume that $(a, b) = g$, so

  $$a = gk, \quad k \in \mathbb{Z}.$$

  In that case,

  $$-a = g(-k).$$

  Since $-k \in \mathbb{Z}$, $g \mid -a$. Trivially, g is still maximal for $(-a, b)$.

  (2). Since all numbers divide 0, and $d \leqslant |a|$ for any number d that divides a, the GCD is therefore $|a|$. ■

- **Relatively prime**: Let $a, b \in \mathbb{Z}$, not both zero. Then, we say that a, b are **relatively prime** if $(a, b) = 1$.

- **Theorem 2.5: *Bézout's identity***: Let a and b be integers, not both 0. Then, there exists integers u, v such that

  $$(a, b) = au + bv.$$

  Furthermore, (a, b) is the smallest positive integer that can be written in this way.

  Proof. Let

  $$S = \{ax + by : \ x, y \in \mathbb{Z},\ ax + by > 0\} \subseteq \mathbb{N}.$$

  First, we show that S has at least one element. If we assume $a \neq 0$, and let $x = a$, $b = 0$, then

  $$a(a) + b(0) = a^2 > 0.$$

  So, $S \neq \varnothing$. By the well-ordering axiom, S has a least element, say

  $$g = au + bv.$$

  If we show that $g = (a, b)$, the proof is complete.

Suppose that a and b has some only common divisor c, then

$$c \mid a \wedge c \mid b.$$

So,

$$c \mid au + bv$$

for all $u, v \in \mathbb{Z}$. Hence, $c \mid g$, and $c \leqslant g$. That is, g would indeed be the greatest common divisor provided we can show that $g \mid a$ and $g \mid b$.

Using the division algorithm,

$$a = gq + r, \quad 0 \leqslant r < g.$$

So,

$$r = a - gq = a - (au + bv)q = a - auq - bvq = a(1 - uq) + b(-vq).$$

Hence, $r \in S$ if $r > 0$. But, $r < g$, contradicting the minimality of g. So, $r = 0$, and $g \mid a$. Using the same argument, we can also get that $g \mid b$. ∎

- **Corollary 2.1**: Let $a, b \in \mathbb{Z}$, $\{a, b\} \neq \{0\}$. Then, a and b are relatively prime if and only if there exists integers u, v such that

$$au + bv = 1.$$

- **Corollary 2.2**: Let $a, b, c \in \mathbb{Z}$ with $\{a, b\} \neq \{0\}$. If $(a, b) = 1$ and $a \mid bc$, then $a \mid c$.

Proof. We have

$$au + bv = 1, \quad u, v \in \mathbb{Z}.$$

So,

$$auc + bvc = c.$$

Since $a \mid a$ and $a \mid bc$,

$$a \mid a(uc) + bc(v),$$

since $uc, v \in \mathbb{Z}$. Thus, $a \mid c$. ∎

- **Corollary 2.3**: Let $a, b \in \mathbb{Z}$, $\{a, b\} \neq \{0\}$. If $(a, b) = 1$, and for same integer n we have $a \mid n$ and $b \mid n$, then $ab \mid n$.

- **Theorem 2.6: Euclidean algorithm**: Let a, b be integers, with b positive. If $b \mid a$, then $(a, b) = b$. Otherwise, apply the division algorithm repeatedly. Let

$$\begin{aligned} a &= bq_1 + r \\ b &= r_1 q_2 + r_2 \\ r_1 &= r_2 q_3 + r_3 \\ &\vdots \\ r_{k-2} &= r_{k-1} q_k + r_k \\ r_{k-1} &= r_k q_{k+1} + 0, \end{aligned}$$

where $q_i, r_i \in \mathbb{Z}$ for all i, and $0 < r_k < r_{k-1} < \cdots < r_1 < b$. Then, $(a, b) = r_k$.

Note: You may have seen this algorithm defined recursively as

$$\gcd(a, b) = \gcd(b, a \bmod b).$$

Proof. If $b \mid a$, then $b \leqslant |a|$. Note that $b \mid b$, so b is trivially the greatest divisor of b. Thus, since $b \leqslant a$ and b is the greatest divisor of b, it should be clear that $(a, b) = b$. Thus, we may assume that $b \nmid a$.

By the division algorithm,

$$a = bq_1 + r_1, \quad q_1, r_1 \in \mathbb{Z}.$$

Notice that $r_1 = a - bq_1$. Assume $c \mid a$ and $c \mid b$. Then,

$$c \mid a(1) + b(-q_1) = a - bq_1 = r_1.$$

Thus, any common divisor of a and b is also a common divisor of b and r_1, with $r_1 = a \bmod b$. Conversely, if $d \mid b$ and $d \mid r_1$, then

$$d \mid b(q_1) + r_1(1) = a.$$

So, any common divisor of b and r_1 is also a common divisor of a and b. Hence, $b, a \bmod b$ have precisely the same common divisors as a and b.

Also, observe that

$$0 < r_1 < b.$$

If $r_1 = 0$, then $b \mid a$, so $r_1 > 0$. Now, we can repeat this argument to say that the common divisors of b, r_1 are the same as the common divisors of r_1, r_2, where $r_2 = b \bmod r_1 = b \bmod a \bmod b$, with

$$0 < r_2 < r_1.$$

Thus,

$$(a, b) = (b, r_1) = (r_1, r_2) = \cdots = (r_{k-1}, r_k) = (r_k, 0) = r_k.$$

Since $0 < r_k < r_{k-1} < \cdots < r_2 < r_1 < b$, each iteration produces a smaller r_i than the one prior, and since all are greater than zero, the sequence must terminate in finite steps.

Note: We note that if $b < 0$, we cannot use the division algorithm, as that requires $b > 0$. This is why we assume b is positive for this theorem. However, recall that

$$(a, b) = (-a, b).$$

This fact, along with $(a, b) = (b, a)$ lets us use the algorithm with

$$(a, -b)$$

to find the gcd of (a, b).

- **Using the Euclidean algorithm**: Note that since

$$r_{k-2} = r_{k-1}q_k + r_k \implies (a, b) = r_k = r_{k-2} - r_{k-1}q_k = r_{k-2}(1) + r_{k-1}(-q_k),$$

with

$$\begin{aligned}
a &= bq_1 + r_1, \\
b &= r_1 q_2 + r_2, \\
r_1 &= r_2 q_3 + r_3, \\
&\vdots \\
r_{k-4} &= r_{k-3} q_{k-2} + r_{k-2}, \\
r_{k-3} &= r_{k-2} q_{k-1} + r_{k-1},
\end{aligned}$$

we can work our way backwards to get integers u, v such that

$$(a, b) = au + bv.$$

For example, consider $a = 45$, $b = 33$. Applying the Euclidean algorithm yields

$$\begin{aligned}
45 &= 33(1) + 12, \\
33 &= 12(2) + 9, \\
12 &= 9(1) + 3, \\
9 &= 3(3) + 0.
\end{aligned}$$

Hence, $(a, b) = 3$. Now, we can find u, v such that $au + bv = 3$. We have

$$\begin{aligned}
3 &= r_1(1) + r_2(-q_3) = 12(1) + 9(-1), \\
&= r_1(1) + (b - r_1 q_2)(-q_3) = 12(1) + (33 - 12(-2))(-1) \\
&= (a - bq_1) + b(-q_3) + r_1 q_2 q_3 \\
&= (a - bq_1) + b(-q_3) + (a - bq_1) q_2 q_3 \\
&= a + b(-q_1) + b(-q_3) + aq_2 q_3 + b(-q_1) q_2 q_3 \\
&= a(1 + q_2 q_3) + b(-q_1 - q_3 - q_1 q_2 q_3) \\
&= 45(1 + 2(1)) + 33(-1 - 1 - 1(2)(1)) = 45(3) + 33(-1 - 1 - 1(2)(1)) = 45(3) + 33(-4).
\end{aligned}$$

Hence, $u = 3$, and $v = -4$.

- **Prime numbers**: A natural number $p > 1$ is said to be prime if its only positive divisors are 1 and p. Otherwise, it is composite. Note that 1 is neither prime nor composite.
- **Euclid's Lemma**: Let $p > 1$ be a positive integer. Then, the following are equivalent.
  1. p is prime; and
  2. if a and b are integers such that $p \mid ab$, then $p \mid a$ or $p \mid b$

Proof. First, suppose that p is prime and $p \mid ab$. Either $(p, a) = 1$ or $(p, a) = p$. If $p \leqslant a$, we have $(p, a) = p$, and so p is a common divisor of both p and a, hence $p \mid a$.

If $p > a$, then $(p, a) = 1$. In that case,

$$pu + av = 1,$$

so

$$pub + avb = b.$$

Now, with $p \mid p$ and $p \mid ab$,

$$p \mid pk + abc, \quad k, c \in \mathbb{Z}.$$

If we let $k = ub$, $c = v$, then

$$p \mid p(ub) + abv = b.$$

So, $p \mid b$. Now, consider the case in which p is composite. In that case, let $p = cd$. Since $p > 1$,

$$1 < c, d < p.$$

So, $p \nmid c$ and $p \nmid d$. ■

- **Corollary 2.4**: Let p be a prime number and $a_1, \cdots, a_n \in \mathbb{Z}$. If

$$p \mid a_1 a_2 \cdots a_n,$$

then $p \mid a_i$ for some i. In fact, every positive integer larger than 1 can be written as a product of primes, called its **prime factorization**.

Proof. Assume $p \in \mathbb{P}$, and $a_1, a_2, \ldots, a_n \in \mathbb{Z}$. Further assume that $p \mid a_1 a_2 \cdots a_n$.

So, $p \mid a_1$ or $p \mid a_2 \cdots a_n$. Further, $p \mid a_2$ or $p \mid a_3 \cdots a_n$. We can extend this idea up to $p \mid a_{n-1}$ or $p \mid a_n$. Thus, $p \mid a_i$ for some i. ■.

- **Theorem 2.8: Fundamental Theorem of Arithmetic**: If $a \in \mathbb{N}$ and $a > 1$, then there exists primes $p_1, \ldots, p_n$ not necessarily distinct such that $a = p_1 p_2 \cdots p_n$. Further, this product is unique up to order. That is, if $a = q_1 q_2 \cdots q_m$ for some primes q_i, then $m = n$, and after rearranging the primes, $p_i = q_i$ for all i.

Proof. Notice that we start with $a = 2$, which is prime. Thus, let $a > 2$ and assume that the theorem is true for smaller numbers. If a is prime, nothing to do. If a is composite, then we can write $a = bc$ for $1 < b, c < a$. Since $b, c < a$, b and c can be written as a product of primes. Thus, a is a product of primes.

Now we prove uniqueness. Suppose that

$$a = p_1 \cdots p_n = q_1 \cdots q_m$$

for some primes p_i and q_i. Without loss of generality, say $n \leqslant m$. Since

$$p_1 \mid a,$$

$p_1 \mid q_i$ for some i. We can rearrange the order to say that $p_1 \mid q_1$. But, q_1 is prime, so $p_1 = 1$ or $p_1 = q_1$. Since 1 is not prime, $p_1 = q_1$. So, we can cancel p_1 from both sides to get

$$p_2 \cdots p_n = q_2 \cdots q_m.$$

We can continue this process, finding that $p_i = q_i$ for $1 \leqslant i \leqslant n$. If $m = n$, we are done. Otherwise, we are left with $1 = q_{n+1} \cdots q_m$. But then $q_m \mid 1$, which is impossible since $q_m > 1$. ■

- **Corollary 2.5**: Let a, b, and n be integers, with $n \neq 0$. If $(a, n) = (b, n) = 1$, then $(ab, n) = 1$.

Proof. Assume that $a, b, n \in \mathbb{Z}$, $n \neq 0$, $(a, n) = (b, n) = 1$.

Assume that $g = (ab, n) > 1$. By the Fundamental Theorem of Arithmetic, g can be factorized as a product of primes. Hence, there exists a prime p that divides g. Since g divides ab and p divides g, $p \mid ab$. In that case, $p \mid a$ or $p \mid b$. But, $p \mid n$ by similar logic. So,

$$(a, n) \geqslant p \quad \text{or} \quad (b, n) \geqslant p.$$

Either way, we have a contradiction, since $(a, n) = (b, n) = 1$. ■.

3.2.3 Properties of the integers

- **Intro**: We note that our discussion will take place in $\mathbb{Z}$, but it could just as easily use $\mathbb{Q}, \mathbb{R}$, or $\mathbb{C}$.

- **Closure**: $\mathbb{Z}$ is closed under addition and multiplication

$$a + b, ab \in \mathbb{Z}$$

  for all $a, b \in \mathbb{Z}$.

- **Associativity**: Addition and multiplication on $\mathbb{Z}$ are both associative.

$$(a + b) + c = a + (b + c),$$
$$(ab)c = a(bc)$$

  for all $a, b, c \in \mathbb{Z}$.

- **Commutative**: Addition and multiplication are both commutative on $\mathbb{Z}$.

$$a + b = b + a,$$
$$ab = ba$$

  for all $a, b \in \mathbb{Z}$.

- **Identity**: We call 0 the additive identity for $\mathbb{Z}$, and 1 the multiplicative identity for $\mathbb{Z}$

$$a + 0 = a, \quad 1a = a$$

  for all $a \in \mathbb{Z}$.

- **Inverse**: If $a \in \mathbb{Z}$, then $-a$ is its additive inverse.

$$a + (-a) = 0.$$

  We note that there is no multiplicative inverse in $\mathbb{Z}$, unlike $\mathbb{Q}, \mathbb{R}$, or $\mathbb{C}$.

3.2.4 Modular arithmetic

- **Intro**: Consider $a, b, n \in \mathbb{Z}$. By the division algorithm,

$$a = nc + r_1,$$
$$b = nk + r_2$$

for $c, k, r_1, r_2 \in \mathbb{Z}$, $0 \leqslant r_1, r_2 < n$. Now, let's consider $a - b$,

$$a - b = nc + r_1 - nk - r_2 = n(c - k) + (r_1 - r_2).$$

Notice that if $n \mid a - b$, then $r_1 - r_2 = 0$, so $r_1 = r_2$. Thus, if $n \mid a - b$, a and b have the same remainder when divided by n.

- **Congruence**: Let $n \geqslant 2$ be an integer. If $a, b \in \mathbb{Z}$, then we say that a is congruent to b modulo n, and write $a \equiv b \pmod{n}$ provided

$$n \mid a - b.$$

That is, a and b have the same remainder when divided by n.

- **Reduction to the remainder**: Suppose that $a, b, n \in \mathbb{Z}$, $n \geqslant 2$, and $n \mid a - b$. So,

$$a \equiv b \pmod{n}.$$

In that case, a and b have the same remainder when divided by n, so

$$a = nc + r,$$
$$b = nk + r$$

for $c, k \in \mathbb{Z}$. Notice

$$a - r = nc,$$
$$b - r = nk.$$

So, $n \mid a - r$, and $n \mid b - r$. Thus, we can reduce to

$$a \equiv r \pmod{n},$$
$$b \equiv r \pmod{n}.$$

- **Theorem 2.9**: Let $n \geqslant 2$ be an integer. Then, $a \equiv b \pmod{n}$ is an equivalence relation on $\mathbb{Z}$. The equivalence class of a consists of all integers having the same remainder as a when divided by n.

Proof. Assume $n \geqslant 2$ is an integer, let $a \equiv b \pmod{n}$ be a relation on $\mathbb{Z}$. First, observe that

$$n \mid 0 = a - a.$$

Thus, $a \equiv a \pmod{n}$, and the relation is reflexive. Next, Assume that

$$a \equiv b \pmod{n}.$$

Thus, $n \mid a - b$, or

$$a - b = nc$$

for some $c \in \mathbb{Z}$. Then,

$$-(b-a) = nc \implies b-a \qquad\qquad\qquad = n(-c).$$

Thus, $n \mid b-a$, so $b \equiv a \pmod{n}$ and the relation is symmetric. Last, assume that

$$a \equiv b \pmod{n}, \quad b \equiv c \pmod{n}.$$

Then,

$$a-b = nr, \quad b-c = nk$$

for some $r, k \in \mathbb{Z}$. Thus,

$$a-b+b-c = a-c = nr+nk = n(r+k).$$

Thus, $n \mid a-c$, so $a \equiv c \pmod{n}$ and the relation is transitive. Hence, we have an equivalence relation on $\mathbb{Z}$.

And, as we know for an equivalence relation $\sim \subseteq S \times S$,

$$[a] = \{b \in S : \ a \sim b\},$$

and if $b \in [a]$, $[a] = [b]$. Thus, for the equivalence relation $a \equiv b \pmod{n}$, let $a \in \mathbb{Z}$, then the equivalence class $[a]$ is

$$[a] = \{b \in \mathbb{Z} : \ a \equiv b \pmod{n}\} = \{a + nk : \ k \in \mathbb{Z}\}.$$

So, the integers $b \in \mathbb{Z}$ with the same remainder when divided by n makes up the equivalence class for a. ■

- **Definition 2.6**: Let $n \geqslant 2$ be an integer. The set of integers modulo n, denoted $\mathbb{Z}_n$ is the set of equivalence classes of $\mathbb{Z}$ with respect to the equivalence relation $a \equiv b \pmod{n}$. We call these the **congruence classes** modulo n. Specifically,

$$\mathbb{Z}_n = \{[0], [1], [2], \ldots, [n-1]\}.$$

  Recall that the representative of a class is not unique. However, it is customary to reduce final answers in $\mathbb{Z}_n$ to the form $[a]$, where $0 \leqslant a < n$.

- **Residue class**: The term **residue class** is another name for **congruence class**. A **residue** is usually a representative of that class.

- **Sumsets**: Consider $S, T \subseteq \mathbb{Z}$. Then,

$$S + T = \{s + t : \ s \in S, t \in T\}$$

  is called the **sumset**. Similarly, we can define the **difference set**, **product set**, or **quotient set** in a similar way, just changing the operation.

- **Addition and multiplication on $\mathbb{Z}_n$**: We have

$$[a] + [b] = [a+b],$$
$$[a][b] = [ab].$$

  Consider the sumset

$$[a] + [b] = \{x + y : \ x \in [a], y \in [b]\}.$$

  Note that

$$[a+b] = \{s \in \mathbb{Z} : \ a+b \equiv s \pmod{n}\}.$$

  So, we are asserting that

$$\{x + y : \ x \in [a], y \in [b]\} = \{s \in \mathbb{Z} : \ a+b \equiv s \pmod{n}\}.$$

Proof. Let $x + y = [a] + [b]$, so $x \in [a]$, and $y \in [b]$. Hence,

$$\begin{aligned} a &\equiv x \pmod{n}, \\ b &\equiv y \pmod{n}. \end{aligned}$$

Thus,

$$a - x = nc, \quad b - y = nk$$

for some $c, k \in \mathbb{Z}$. Thus,

$$a - x + b - y = a + b - (x + y) = n(c + k).$$

Hence,

$$a + b \equiv x + y \pmod{n}.$$

So, $x + y \in [a + b]$, and $[a] + [b] \subseteq [a + b]$. Now, let $z \in [a + b]$, so

$$z \equiv a + b \pmod{n}.$$

Let $x = a$, and $y = z - a$. Then, $x + y = z$,

$$x \equiv a \pmod{n},$$

and

$$z - a = y \equiv (a + b) - a \pmod{n}.$$

So, $y \equiv b \pmod{n}$, hence $x \in [a]$ and $y \in [b]$. Therefore $[a + b] \subseteq [a] + [b]$, and we can conclude that

$$[a] + [b] = [a + b].$$

■

Now, consider

$$\begin{aligned} [a][b] &= \{xy : \ x \in [a], y \in [b]\}, \\ [a \cdot b] &= \{z \in \mathbb{Z} : \ ab \equiv z \pmod{n}\}. \end{aligned}$$

We assert that

$$\{xy : \ x \in [a], y \in [b]\} = \{z \in \mathbb{Z} : \ ab \equiv z \pmod{n}\}.$$

The proof for this is non-trivial. Instead of proving these two operations, we can instead define them as such, and show that they are well-defined.

- **A well-defined operation on a set**: An operation is well-defined on a set if it gives a single, unambiguous output in that set for every allowed input from that set. For a binary operation $*$ on a set S,

$$* : \ S \times S \to S.$$

$*$ is well-defined on S if for every $a, b \in S$, $a * b$ exists, is unique, and is also an element of S.

- **Checking well-definedness**: You must show well-definedness whenever your definition uses representatives of equivalence classes. As well as for defining operations on a set.

In general, whenever you define an operation on a set, you should at least know that it is well-defined. You should check well-definedness especially when the rule involves choices, representatives, cases, or possible ambiguity.

- **Theorem 2.10**: For any integer $n \geqslant 2$, addition and multiplication on $\mathbb{Z}_n$ are well-defined.

  Proof. Suppose that $a_1 \equiv a_2 \pmod{n}$ and $b_1 \equiv b_n \pmod{n}$. Then,

  $$a_1 - a_2 = nc,$$
  $$b_1 - b_2 = nk.$$

  So,

  $$a_1 + b_1 - (a_2 + b_2) = n(c + k).$$

  Thus,

  $$a_1 + b_1 \equiv a_2 + b_2 \pmod{n}.$$

  Hence, $[a_1 + b_1] = [a_2 + b_2]$, and we can say that addition is well-defined.

  Now,

  $$a_1 = nc + a_2,$$
  $$b_1 = nk + b_2.$$

  So,

  $$a_1b_1 = (nc + a_2)(nk + b_2) = n^2ck + ncb_2 + nka_2 + a_2b_2.$$

  Hence,

  $$a_1b_1 - a_2b_2 = n(nck + cb_2 + ka_2),$$

  and we can say that $a_1b_1 \equiv a_2b_2$, so

  $$[a_1b_1] = [a_2b_2].$$

  Therefore, multiplication on residue classes is also well-defined.

- **Theorem 2.11**: Let $n \geqslant 2$ be an integer, and take any $[a], [b], [c] \in \mathbb{Z}_n$. Then,
  1. $[a] + [b] \in \mathbb{Z}_n$
  2. $[a] + ([b] + [c]) = ([a] + [b]) + [c]$
  3. $[a] + [b] = [b] + [a]$
  4. $[a] + [0] = [a]$
  5. $[a] + [-a] = [0]$

- **Theorem 2.12**: Let $n \geqslant 2$ be an integer, and $[a], [b], [c] \in \mathbb{Z}_n$, then
  1. $[a][b] \in \mathbb{Z}_n$
  2. $[a]([b][c]) = ([a][b])[c]$
  3. $[a][b] = [b][a]$
  4. $[a]([b] + [c]) = [a][b] + [a][c]$
  5. $[1][a] = [a]$

- **Notational switch**: When working in $\mathbb{Z}_n$, we will normally simply write a or $a + b$ instead of $[a]$ or $[a] + [b]$, as long as the context is clear.

3.3 Groups

3.3.1 Introduction to groups

- **Permutations**: Let A be the set $\{1, 2, 3\}$. We would like to consider all the permutations of A. The notation we will use is

$$\sigma = \begin{pmatrix} 1 & 2 & 3 \\ a & b & c \end{pmatrix},$$

  which means $\sigma(1) = a, \sigma(2) = b$, and $\sigma(3) = c$.

- **Composition of permutations**: Consider

$$\sigma = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}, \quad \tau = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}.$$

  Then,

$$\begin{aligned} (\sigma \circ \tau)(1) &= \sigma(\tau(1)) = \sigma(3) = 3, \\ (\sigma \circ \tau)(2) &= \sigma(\tau(2)) = \sigma(1) = 2, \\ (\sigma \circ \tau)(3) &= \sigma(\tau(3)) = \sigma(2) = 1. \end{aligned}$$

  Thus,

$$\sigma \circ \tau = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}.$$

- **Properties of permutations**:

  1. **Closure**: Permutations are closed under composition
  2. **Associativity**: For permutations σ, τ, ρ, we have $\rho \circ (\sigma \circ \tau) = (\rho \circ \sigma) \circ \tau$
  3. **Identity**: If σ is any permutation of A, then

$$\sigma \circ \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} \circ \sigma = \sigma.$$

  4. **Inverse**: For each permutation σ, there is another permutation τ such that

$$\sigma \circ \tau = \tau \circ \sigma = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}.$$

  Note that we do not have commutativity, $\sigma \circ \tau \neq \tau \circ \sigma$ in general.

- **Symmetric group on n letters, S_n**: Permutations under composition give us an example of a group. If we let $A = \{1, 2, 3, \ldots, n\}$, for any positive integer n, then we call the set of all permutations of this set, under the composition operation the **symmetric group on n letters**, denoted S_n.

- **Groups**: A **group** is a set G, together with a binary operation $*$, satisfying the following conditions

  1. $a * b \in G$ for all $a, b \in G$ (closure);
  2. $(a * b) * c = a * (b * c)$ for all $a, b, c \in G$ (associativity);
  3. there exists an $e \in G$ such that $a * e = e * a = a$ for all $a \in G$ (existence of identity); and
  4. for each $a \in G$, there exists a $b \in G$ such that $a * b = b * a = e$ (existence of inverses).

  We refer to G as a group **under** $*$. The element e is called the **identity** of the group. Also, if b is the inverse of a, then we write $b = a^{-1}$.

- **Abelian group**: A group G is said to be **abelian** if $a * b = b * a$ for all $a, b \in G$. So, $*$ is commutative.

Note that $\mathbb{Z}$ and $\mathbb{Z}_n$ for $n \geqslant 2$ are abelian groups under addition. In fact, 0 is the identity, and the inverse of a is $-a$. The same can be said for $\mathbb{Q}, \mathbb{R}$, and $\mathbb{C}$ under addition.

- **Group tables**: When a group G has only finitely many elements, we can represent it with a **group table**. We write the elements of G down the first column and along the first row of the table. Then, the entry in the row headed by a and the column headed by b is $a * b$. Observe the group table for the additive group $\mathbb{Z}_5$

	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

- **Example**: Let G be the set of nonzero rational numbers. Then, G is an abelian group under multiplication.

Proof. Let

$$G = \{q \in \mathbb{Q} : \; q \neq 0\} \subseteq \mathbb{Q}.$$

Consider $q_1, q_2, q_3 \in G$, with

$$q_1 = \frac{a}{b}, \; q_2 = \frac{c}{d}, \; q_3 = \frac{e}{f}.$$

Then,

$$q_1 q_2 = \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}.$$

It should be clear that $a, b, c, d \neq 0$, so $q_1 q_2 \in G$. Thus, we have closure. Next,

$$q_1 \cdot (q_2 \cdot q_3) = (q_1 \cdot q_2) \cdot q_3$$

since multiplication is associative. Also,

$$q_1 q_2 = q_2 q_1$$

since multiplication is commutative. Clearly $e = 1$ is the identity, since $1 \in G$ and $1q = q$ for all $q \in G$. Also, for

$$q = \frac{a}{b} \in G,$$

$$q \cdot \frac{1}{q} = \frac{a}{b} \cdot \frac{b}{a} = 1.$$

Thus, $q^{-1} = \frac{1}{q}$.

- **Example**: Let $n \geqslant 2$ be positive integer. Let $U(n)$ denote the set of all elements $a \in \mathbb{Z}_n$ such that $(a, n) = 1$.

So,

$$U(n) = \{a \in \mathbb{Z}_n : \ (a, n) = 1\}.$$

First, lets check that the condition $(a, n) = 1$ is well-defined. Thus, if

$$a \equiv b \pmod n$$

with $(a, n) = 1$, we should see that $(b, n) = 1$. Well, we have

$$\begin{aligned} a &= b + nk, \\ au + nv &= 1. \end{aligned}$$

So,

$$(b + nk)u + nv = bu + nku + nv = bu + n(ku + v) = 1.$$

Thus, $(b, n) = 1$. Alternately, since a is a linear combination of b and n, then if we let c be a divisor of b and n, we can invoke lemma 2.1 to get that

$$c \mid b(1) + nk.$$

So, $c \mid a$. Now, c divides both a and n, but the only common divisor for a and n is 1. Thus, $c = 1$. Now, notice that we choose an arbitrary divisor for b and n, and deduced that it must be one. Thus, b and n have no common divisor other than 1, so

$$(b, n) = 1.$$

$\blacksquare$.

Now that we have well-definedness, we can claim that $U(n)$ forms an abelian group under multiplication in $\mathbb{Z}_n$.

First, we check for closure. If $a, b \in U(n)$, then we should see that $ab \in U(n)$. So, we require that

$$(ab, n) = 1.$$

In fact, we see this to be true by Corollary 2.5. We also know multiplication in $\mathbb{Z}_n$ to be both associative and commutative, and $1 \in U(n)$ is the identity. But, what about inverses? For a residue class a, we call $[b] = a^{-1}$ if

$$ab \equiv 1 \pmod n.$$

In other words,

$$ab = 1 + nk, \ k \in \mathbb{Z}.$$

Notice that since $(a, n) = 1$,

$$au + nv = 1 \implies au = 1 - nv = 1 + nk.$$

Thus, $u = a^{-1}$.

- **Example**: Let n be a positive integer, and

$$(G, \cdot) = (\{w \in \mathbb{C} : \ w^n = 1\}, \cdot).$$

If $w, z \in G$, then

$$(wz)^n = w^n z^n = 1(1) = 1.$$

So, we have closure. Next, we know that multiplication is both associative and commutative, and $1 \in G$ is the identity. Now, for $w \in G$, $\frac{1}{w} \in \mathbb{C}$, and

$$\left(\frac{1}{w}\right)^n = \frac{1}{w^n} = \frac{1}{1} = 1.$$

So, $\frac{1}{w} \in G$, and $\frac{1}{w} = w^{-1}$, since

$$w\left(\frac{1}{w}\right) = 1.$$

- **Trivial group**: We call the group that consists of just the identity element the **trivial group**.

- **General linear group**: The set of all non-singular matrices in $M_n(\mathbb{R})$ ($\mathbb{R}^{n\times n}$) is called the **general linear group** over $\mathbb{R}$, and denoted $GL_n(\mathbb{R})$. It is a group under matrix multiplication. The identity matrix I_n is the identity of $GL_n(\mathbb{R})$. Also, if $A, B \in GL_n(\mathbb{R})$, then

  $$AB(B^{-1}A^{-1}) = A(BB^{-1})A = I_n,$$

  and

  $$(B^{-1}A^{-1})AB = I_n.$$

  So, we have closure. Also, we know that matrix multiplication is associative, and by definition every element of A has an inverse. Notice that

  $$(A^{-1})^{-1} = A,$$

  so $A^{-1} \in GL_n(\mathbb{R})$. Thus, this is indeed a group. However, we know that matrix multiplication is not generally commutative, so the group is nonabelian.

- **Definition 3.3 (*direct product*)**: Let G be a group with operation $*$ and H group with operation $\bullet$. On the Cartesian product $G \times H$, define an operation $\diamond$ via

  $$(g_1, h_1) \diamond (g_2 h_2) = (g_1 * g_2, h_1 \bullet h_2)$$

  for all $g_i \in G$, $h_i \in H$. Under this operation, we call $G \times H$ the **direct product** of G and H.

  Note: So, the underlying set is the Cartesian product, with operation $\diamond$. Thus, the group is

  $$(G \times H, \diamond).$$

- **Theorem 3.1**: The direct product of two groups is a group.

  Proof. Assume that $(G, *)$ and $(H, \bullet)$ are groups. Then, let $(G \times H, \diamond)$ be the direct product.

If $a_1, a_2 \in G \times H$, then $a_1 = (g_1, h_1)$, and $a_2 = (g_2, h_2)$. Now,

$$a_1 \diamond a_2 = (g_1, h_1) \diamond (g_2, h_2) = (g_1 * g_2, h_1 \bullet h_2).$$

Since G, H are groups, $g_1 * g_2 \in G$, and $h_1 \bullet h_2 \in H$, so

$$(g_1 * g_2, h_1 \bullet h_2) \in G \times H.$$

Thus, we have closure. Now, let $a_1, a_2, a_3 \in G \times H$, we have

$$\begin{aligned}((g_1, h_1) \diamond (g_2, h_2)) \diamond (g_3, h_3) &= (g_1 * g_2, h_1 \bullet h_2) \diamond (g_3, h_3)\\ &= (g_1 * g_2 * g_3, h_1 \bullet h_2 \bullet h_3),\end{aligned}$$

with

$$\begin{aligned}(g_1, h_1) \diamond ((g_2, h_2) \diamond (g_3, h_3)) &= (g_1, h_1) \diamond (g_2 * g_3, h_2 \bullet h_3)\\ &= (g_1 * g_2 * g_3, h_1 \bullet h_2 \bullet h_3).\end{aligned}$$

So, we have associativity. If $e_G \in G$, and $e_H \in H$ are the identities for G and H, then its painfully obvious that

$$(e_G, e_H)$$

is the identity for $G \times H$. Furthermore, (g^{-1}, h^{-1}) is the inverse. ■

3.3.2 A few basic properties

- **Notation change**: From now on, when working inside a group, we will suppress the symbol for the group operation. So, we will write ab instead of $a * b$. Although, if the operation is addition, the **multiplicative notation** would be confusing. In that case, we will use the **additive notation**, and continue with $a + b$

  Further, if the operation is addition, we will write 0 instead of e, and $-a$ instead of a^{-1}.

- **Theorem 3.2**: Let G be a group. Then,

  1. the identity of G is unique, and
  2. if $a \in G$, then a^{-1} is unique.

  Proof. (1) Suppose that G is a group, and e, f are identities in G. Then,

  $$\begin{aligned} ef &= f, \\ ef &= e. \end{aligned}$$

  Thus, $e = f$. Now, suppose that for $a \in G$, b, c are both inverses for a. Then,

  $$\begin{aligned} bac &= (ba)c = ec = c, \\ bac &= b(ac) = be = b. \end{aligned}$$

  Thus, $c = b$.

- **Theorem 3.3**: In any group, $(ab)c = a(bc)$. Thus, we can write abc without any ambiguity. But, we would like to be able to write $abcd$, for instance. This brings us to the following theorem

  Let G be any group, and $a_1, a_2, \ldots, a_n \in G$. Then, regardless of how the product $a_1 a_2 \cdots a_n$ is bracketed, the result equals

  $$(\cdots(((a_1 a_2)a_3)a_4)\cdots a_{n-1})a_n.$$

  Proof. We proceed by strong induction upon n. If n is 1 or 2, no bracketing is needed. If $n = 3$, we have associativity from the definition of a group. So, let $n \geqslant 4$. Assume that the theorem is true for any product of fewer than n group elements.

  Let

  $$w = a_1 \cdots a_n$$

  be a product with any bracketing configuration. Then, let

  $$w = xy$$

  for $x = a_1 \cdots a_m$, $y = a_{m+1} \cdots a_n$, each with some bracketing. By the inductive hypothesis,

  $$\begin{aligned} x &= (\cdots((a_1 a_2)a_3)\cdots a_{m-1})a_m, \\ y &= (\cdots((a_{m+1} a_{m+2})a_{m+3})\cdots a_{n-1})a_n. \end{aligned}$$

  So,

  $$w = xy = ((\cdots((a_1 a_2)a_3)\cdots a_m)(\cdots((a_{m+1} a_{m+2})a_{m+3})\cdots a_{n-1}))a_n.$$

  Applying our inductive hypothesis to the product of the first $n - 1$ terms, we obtain the desired bracketing. ■

- **Theorem 3.4**: Let G be a group, with $a, b \in G$. Then,

  1. $(a^{-1})^{-1} = a$
  2. $(ab)^{-1} = b^{-1}a^{-1}$

  Proof. Since

  $$aa^{-1} = a^{-1}a = e,$$

  $(a^{-1})^{-1} = a$ by definition of the inverse. Next,

  $$abb^{-1}a^{-1} = aea^{-1} = aa^{-1} = e,$$
  $$b^{-1}a^{-1}ab = b^{-1}eb = b^{-1}b = e.$$

  Thus, $(ab)^{-1} = b^{-1}a^{-1}$. ■

- **Theorem 3.5 (*cancellation law*):** Let G be a group and $a, b, c \in G$. If either $ab = ac$ or $ba = ca$, then $b = c$.

  Proof. Let $ab = ac$, then

  $$a^{-1}ab = aa^{-1}ac \implies b = c.$$

  Similarly, if $ba = ca$, then

  $$baa^{-1} = caa^{-1} \implies b = c.$$

  ■

- **Corollary 3.1**: Let G be a group, and $a, b \in G$. Then, there exists exactly one $c \in G$ such that $ac = b$, and exactly one $d \in G$ such that $da = b$.

  Proof. Let $a, b \in G$. Assume that there exists two elements $c, f \in G$ such that

  $$ac = af = b.$$

  Then, by Theorem 3.5,

  $$ac = af \implies c = f.$$

  Now, assume that $d, f \in G$ such that

  $$da = fa = b.$$

  Then,

  $$da = fa \implies d = f.$$

  ■

3.3.3 Powers and orders

- **Definition 3.4**: If G is a group, then its **order**, $|G|$, is the number of elements in the set G. We say that G is a **finite group** if its order is finite; otherwise it is an **infinite group**.

- **Powers of group elements**: Let G be any group, and $a \in G$. Then, for any positive integer n, we let

$$a^n = \underbrace{aa \cdots a}_{n \text{ times}}.$$